

EDUCATION

Degree/Certificate	Institute/School	CGPA/Percentage	Year
B.Tech. EE	Indian Institute of Technology, Jodhpur	7.38/10	2024-Present
Senior Secondary	Tapovan School, Rajkot	88.8%	2024
Secondary	Tapovan School, Rajkot	90.8%	2022

SELECTED TECHNICAL HIGHLIGHTS

Software Engineering: Python/Rust engineer with production-style backend, systems, and developer-tooling projects: CORTEX has 180+ FastAPI endpoints, 10 domain routers, and 1,900+ automated tests; ASTERA supports 14 languages with 153 tests and 92 Criterion benchmarks.

Systems & ML Outcomes: Implemented agent run loops, hybrid retrieval, Tree-sitter static analysis, concurrent blockchain validation, PPO traffic-control training over 500,000+ timesteps, and Jetson inference at 10–14 FPS.

Target Roles: Backend engineering, infrastructure, AI tooling, static analysis, ML systems, and systems programming roles requiring APIs, databases, concurrency, testing, and reproducible evaluation.

EXPERIENCE

- Undergraduate Researcher - Computer Vision, IIT Jodhpur

| *May 2026 – Present* | [GitHub](#)

Hybrid CNN–Attention Architectures for Robust Classification of Imbalanced Natural Image Datasets

PyTorch, Computer Vision, Medical Image Analysis, Reproducible ML Evaluation

- Researching imbalanced image classification under [Prof. Pratik Mazumder](#); designed experiments to measure robustness across class imbalance, disease categories, and real-world image variation.
 - Reproduced PLDNet as a controlled baseline and fixed dataset splits, training configuration, evaluation scripts, and comparison protocol before testing CNN–Attention variants.
 - Evaluated 30+ model variants with ablations on attention placement, feature fusion, and representation changes; tracked per-class behavior instead of relying only on aggregate accuracy.
 - Benchmarked Mendeley Potato Leaf Disease (3,076 images, 7 imbalanced classes) and curated PlantVillage data (2,152 images, 3 classes) using reproducible loaders, baselines, and evaluation artifacts.
 - Prepared reporting pipeline for accuracy, F1 score, per-class recall, confusion matrices, and PLDNet-relative comparisons after final experimental review.
- Open Source Contributor - plink, Devlup Labs, IIT Jodhpur

| *May 2025 – Nov. 2025* | [GitHub](#)

Secure Peer-to-Peer File Transfer across Heterogeneous Networks

Python, TCP/IP, Sockets, Multithreading, AES-256, NAT Traversal, Git

- Contributed networking modules for a peer-to-peer file-transfer system supporting direct connections, UPnP/NAT traversal research, role reversal, and heterogeneous client networks.
 - Implemented TCP socket paths for encrypted file chunks, metadata exchange, multithreaded transfer coordination, parsing refactors, direct-connection management, and port-scanning utilities.
 - Reviewed protocol behavior, debugged connection failures, documented setup and transfer flows, and coordinated changes through Git issues, pull requests, and integration testing.
 - Completed 16 assigned tasks across cryptography, networking research, file metadata, chunk handling, parsing refactors, port scanning, UDP hole-punching prototypes, network-type detection, and documentation.

FLAGSHIP PROJECTS

- CORTEX

| *May 2026 – Present* | [GitHub](#)

Local-First AI Infrastructure for Developer Intelligence

Python, FastAPI, Next.js, React, TypeScript, PostgreSQL, Redis, Qdrant, JWT, Agent Systems

- Designed a local-first AI infrastructure platform using FastAPI, Next.js 15, React 19, PostgreSQL, Redis, embedded Qdrant, arq workers, JWT/CSRF auth, Argon2, Fernet/PBKDF2 encryption, and 180+ endpoints across 10 routers.
 - Organized the backend into 106 services, 44 models, 45 schemas, and 37 migrations for memory, privacy, awareness, interaction, integration, cognition, developer intelligence, and health workflows.
 - Implemented agent execution loop components for tool dispatch, output verification, stall detection, context compaction, token accounting, event handling, and policy checks.
 - Implemented hybrid retrieval over episodic, semantic, and working memory with vector search, knowledge-graph recall, repository analysis, file tracking, GitHub sync, model catalogs, providers, and recommendations.
 - Validated agents, REST APIs, memory, privacy, services, and infrastructure with 1,900+ automated tests across 10 routers; maintained documentation, ADRs, privacy controls, and frontend quality gates.
- ASTERA

| *Jun. 2026 – Present* | [GitHub](#)

Production-Grade Static Analysis & Code Intelligence Engine

Rust, Tree-sitter, SQLite FTS5, Axum, Tokio, Rayon, Static Analysis, REST APIs, Semantic Search

- Built a local-first static-analysis engine that parses repositories into a queryable Code Property Graph for symbol lookup, semantic search, dependency traversal, impact analysis, metrics, and architecture checks.
 - Implemented a 12-crate Rust workspace with Tree-sitter parsers for 14 languages, Axum REST APIs, WebSocket events, plugin hooks, SQLite FTS5 indexing, CLI commands, and incremental file watching.
 - Covered parser, extractor, storage, REST API, search, graph traversal, and metrics paths with 153 automated tests and 92 Criterion benchmarks across 11 suites.
 - Optimized hot paths to approximately 33 ns language classification, 3.27 μs API statistics responses, 16.16 μs SQLite FTS5 symbol search, and 1 μs symbol/edge count queries.
 - Benchmarked repository analysis at 106–110 μs small-file parsing, 2–5 ms medium-repository parsing, 5.08–12.69 ms Parse–Extract–Store latency, 3–24 μs impact traversals, and 229 ms metrics over 10,000 graph nodes.

•Jetson-Enabled UAV Systems Integration

| Aug. 2025 – Mar. 2026 | [GitHub](#)

Companion-Computer Architecture for MAVLink Control, Telemetry, and Onboard Perception

Python, NVIDIA Jetson, MAVLink, Cube Orange, ArduPilot, YOLO, Embedded Linux

- Integrated an NVIDIA Jetson companion computer with a Cube Orange flight controller over MAVLink for command exchange, telemetry streaming, serial debugging, and onboard perception workflows.
- Assembled and manually flight-tested a quadcopter platform under [Dr. Hardik Jain](#), validating approximately 20 minutes of endurance on a 3S lithium-polymer battery.
- Implemented MAVLink communication scripts for telemetry inspection, mode switching, arming, takeoff sequencing, altitude monitoring, and landing checks.
- Deployed YOLO-based object detection on Jetson and measured approximately 10–14 FPS onboard inference while validating communication and perception subsystems during flight tests.
- Documented subsystem integration, flight-test observations, and handoff steps for a long-term UAV research platform.

•Parallel Blockchain Simulator

| Nov. 2025 – Dec. 2025 | [GitHub](#)

Concurrent Blockchain Simulation with Thread-Safe Validation

Python, Multithreading, SHA-256, Queues, Hash Maps, Merkle Trees, Mutex Locks

- Coordinated 7 blockchain threads—3 miners, 3 validators, and 1 transaction generator—using mutex locks and events around shared chain state.
- Modeled account balances, linked-list blocks, queued transactions, hash-map mempool lookup, Merkle verification, and SHA-256 Proof-of-Work at difficulty 3.
- Stress-tested synchronization under randomized loads, processing 131–200 validated transactions per simulation and averaging approximately 174 transactions across runs.
- Generated 20–59 mined blocks per simulation at difficulty 3, averaging approximately 46 blocks per run without corrupting chain state.

•Adaptive Traffic Signal Control using Reinforcement Learning

| May 2026 | [GitHub](#)

PPO-Based Traffic-Control Optimization

Python, PyTorch, Stable-Baselines3, Gymnasium, TensorBoard

- Trained PPO traffic-signal policies for 500,000+ timesteps using 16 vectorized Gymnasium environments and TensorBoard diagnostics.
- Defined state and reward features for vehicle density, waiting time, stochastic arrivals, congestion penalties, and throughput in a custom RL environment.
- Measured policy improvement over random-control baselines with 2341.5 final test reward and 2371.47 mean evaluation reward (25.57 standard deviation).

ADDITIONAL ENGINEERING PROJECTS

•Edge AI Face Recognition System

| Feb. 2025 | [Project](#)

Offline Face Recognition on ESP32-CAM

ESP32-CAM, Edge Impulse, Arduino IDE, C++, TinyML

- Deployed offline face recognition on ESP32-CAM with Edge Impulse, achieving approximately 90% internal evaluation accuracy through on-device inference.
- Collected a custom dataset of approximately 100–150 images across 5 individuals, with 20–30 images per person across multiple angles and orientations.
- Completed the TinyML workflow from data collection and labeling to model training, Arduino IDE deployment, embedded inference, and offline recognition.

•Game Centre

| Mar. 2025 – Apr. 2025 | [GitHub](#)

Console-Based Multi-Game Platform with User Authentication

C, Modular Programming, File I/O, Git

- Led a 5-member team building a menu-driven C gaming platform with 3 classic games and shared module interfaces.
- Implemented Minesweeper, authentication, login validation, persistent sessions, saved state, and structured file I/O.
- Managed the central GitHub repository, resolved merge conflicts, coordinated module integration, and led testing/debugging before final submission.

•Autonomous Wall-Following Robot

| Jan. 2025 | [GitHub](#)

PID-Controlled Differential-Drive Robot

Arduino Uno/Nano, C++, HC-SR04 Sensors, L298N Motor Driver

- Programmed autonomous wall-following navigation on Arduino Uno/Nano using 3 HC-SR04 ultrasonic sensors and an L298N differential-drive motor driver.
- Implemented PID control with integral windup protection, output limiting, dynamic derivative calculation, turn reset logic, and serial diagnostics for tuning.
- Used ultrasonic feedback for obstacle avoidance, left/right wall selection, autonomous turns, and continuous motor speed correction.

TECHNICAL SKILLS

Languages: Python, Rust, C, C++, TypeScript, JavaScript, SQL, Embedded C/C++

Backend & APIs: FastAPI, Axum, REST APIs, WebSockets, Pydantic, SQLAlchemy, Alembic, JWT, CSRF, Authentication, Rate Limiting, Microservices

Systems: Data Structures, Algorithms, Operating Systems, Computer Networks, Concurrency, Multithreading, Synchronization, Async Programming, Tokio, Rayon, Mutexes, Events, IPC, CLI Development

Databases & Search: PostgreSQL, SQLite, SQLite FTS5, Redis, Qdrant, pgvector, Vector Search, Semantic Search, Hybrid Search, Full-Text Search, BM25, HNSW

AI / ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, YOLO, TensorBoard, CNNs, Transformers, Attention, Feature Fusion, Class-Imbalance Evaluation

Developer Tools: Tree-sitter, Code Property Graphs, Static Analysis, Symbol Resolution, Dependency Analysis, Impact Analysis, Program Analysis, Git, GitHub, pytest, Criterion

Robotics & Embedded: NVIDIA Jetson, Jetson Orin NX, Cube Orange, ArduPilot, Mission Planner, DroneKit, MAVLink, ESP32-CAM, Arduino, Raspberry Pi, Sensor Integration, TinyML, Edge AI

DevOps & Frontend: Linux, Docker, Docker Compose, CI/CD, GitHub Actions, Jupyter, MATLAB, Serde, Clap, Notify, Tracing, React, Next.js, TypeScript, Tailwind CSS, Figma

RELEVANT COURSEWORK

- Data Structures and Algorithms; Pattern Recognition and Machine Learning; Probability, Statistics and Stochastic Processes; Operating Systems; Computer Networks; Signals & Systems; Digital Design; Introduction to Computer Science

ACHIEVEMENTS

- Secured All India Rank (AIR) 7,932 in JEE Main 2024 among 1,415,110 candidates nationwide.
- Secured All India Rank (AIR) 8,663 in JEE Advanced 2024 among 180,200 candidates, earning admission to the Indian Institute of Technology (IIT) Jodhpur.

LEADERSHIP & CAMPUS INVOLVEMENT

Secretary , Designerds – The Design Club, IIT Jodhpur	<i>Aug. 2025 – Apr. 2026</i>
• Guided 50+ students, ran design workshops, and helped secure 7th rank at Inter IIT Cultural Meet 8.0.	
Student Guide , Student Wellbeing Committee (SWC), IIT Jodhpur	<i>Aug. 2025 – Apr. 2026</i>
• Supported 12 first-year students through academic transition and campus onboarding.	
Core Member , Inside – The Game Development Society, IIT Jodhpur	<i>Aug. 2025 – Apr. 2026</i>
• Coached 5 first-year students on an Inside-backed game-building project.	
Assistant Head , Ignus’26 – IIT Jodhpur	<i>Jan. 2026 – Feb. 2026</i>
• Led 30+ volunteers and managed digital design for Ignus’26.	
Assistant Head , Prometeo’26 – IIT Jodhpur	<i>Oct. 2025 – Jan. 2026</i>
• Led 30+ volunteers and managed digital design for Prometeo’26.	
Assistant Head , Varchas’25 – IIT Jodhpur	<i>Sep. 2025 – Nov. 2025</i>
• Led 30+ volunteers and managed digital design for Varchas’25.	